

Combinatorial normal boundary intersection with spectral screening for structurally dependent many-objective optimization

Gabriel Victor de Lima¹, Mirelli de Castro Cesário¹, Anderson Paulo de Paiva¹

¹ Institute of Industrial Engineering and Management, Federal University of Itajubá, Itajubá, Brazil

d25101099@unifei.edu.br, mirellicesario12@gmail.com, andersonppaiva@unifei.edu.br

ABSTRACT

Engineering decision problems often involve many dependent objectives, whereas classical normal boundary intersection becomes dimensionally inadmissible when the number of objectives exceeds the number of decision variables plus one. This study proposes combinatorial normal boundary intersection (CNBI), which diagnoses the effective objective structure, screens geometrically informative objective subsets, solves admissible lower-dimensional normal boundary intersection subproblems, and recomposes all candidates in the original objective space. A controlled synthetic experiment varied decision-space dimension, dimensional excess, and structural dependence across 27 scenarios with 10 replications. CNBI was compared with varimax rotated-factor normal boundary intersection and two established evolutionary algorithms. Using the complete fronts produced by each method, CNBI achieved the best inverted generational distance in 26 scenarios and the best hypervolume in 22. Fixed-cardinality curves showed that CNBI retained the best median coverage from 5 to 20 points. A nine-scenario ablation reduced evaluations by a median of 70.5%, with limited median losses in coverage and hypervolume. In an eight-response turning application, CNBI obtained an inverted generational distance of 0.0754 and a hypervolume of 0.5023, remaining competitive with the strongest evolutionary comparator. Tests on MaF8 and MaF9 supported generalization, whereas MaF13 exposed a limitation involving repeated decisions and extreme solutions. The results indicate that spectral screening can make normal-boundary search practical for structurally dependent many-objective response-surface problems while preserving the meaning of the original objectives.

Keywords: Multi-objective optimization; normal boundary intersection; spectral screening; parallel analysis; response surface methodology; manufacturing decision support

Highlights:

- CNBI extends normal boundary intersection to structurally dependent objectives.
- Spectral screening selects admissible subsets without discarding objectives.
- CNBI attains the best complete-front IGD in 26 of 27 synthetic scenarios.
- Screening cuts median evaluations by 70.5% in a nine-scenario ablation.
- An eight-response turning case confirms practical decision-support value.

1. Introduction

Engineering decision problems increasingly require the simultaneous optimization of multiple performance criteria. In manufacturing, for example, quality, productivity, reliability, resource consumption, and economic performance may depend on the same set of decision variables while representing correlated and conflicting objectives (Chiao & Hamada, 2001; Tong et al., 2005). A machining condition that extends tool life, for instance, may reduce productivity or increase cost. The resulting task is therefore not to identify a single optimum, but to construct a set of efficient alternatives that exposes the trade-offs available to engineers and decision makers (Jeong et al., 2024).

This challenge becomes particularly relevant when a relatively small number of decision variables gives rise to many objectives. Such a structure is common in engineering optimization and is especially frequent in experimental process modeling, where several performance responses are obtained from the same controllable factors. Design of experiments (DOE) and response surface methodology (RSM) provide a systematic basis for modeling this type of problem (Kumar & Reji, 2023). Designed experiments organize data collection, whereas response surfaces translate the observed effects of controllable factors into explicit mathematical models within a defined experimental region (Montgomery, 2013; Myers et al., 2016). Second-order models are particularly useful because they can represent linear effects, curvature, and interactions using a relatively small number of experiments. Nevertheless, DOE and RSM constitute only one important class of applications: the underlying optimization problem is more general and arises whenever the number and dependence structure of the objectives challenge conventional multiobjective formulations.

Normal boundary intersection (NBI) is attractive in this context because it provides an explicit geometric mechanism for generating Pareto-efficient alternatives. Starting from the individual objective optima, NBI constructs the convex hull of individual minima (CHIM) and generates search directions normal to that hull (Das & Dennis, 1998). Compared with simple aggregation schemes such as weighted sums, this construction can provide a more uniform representation of the Pareto frontier and recover solutions in nonconvex regions that may otherwise be missed (Das & Dennis, 1997). NBI has consequently been incorporated into multiresponse optimization and continues to be investigated in manufacturing decision problems, including recent process-planning applications combining NBI with exact and metaheuristic search strategies (Mechaacha et al., 2026; Moura et al., 2018).

The direct NBI formulation, however, has a structural dimensional limitation. For a fixed CHIM coefficient vector, an NBI subproblem relates the m objective-space coordinates through the decision vector $x \in \mathbb{R}^{n_x}$ and a scalar displacement variable. Consequently, when $m > n_x + 1$, the formulation contains more objective-space equality equations than free search variables and is generically overdetermined. This condition is determined by the dimensions of the objective and decision spaces rather than by statistical correlation and can therefore occur even when the objectives are mutually independent. More generally, increasing the number of objectives changes the structure and difficulty of multiobjective optimization problems, and is particularly relevant when many performance criteria are described using a comparatively small number of decision variables (Allmendinger et al., 2022).

Dependence among objectives creates a second, distinct difficulty. Redundant or structurally dependent objectives can reduce the effective dimensionality of the objective space and produce linearly dependent or nearly collinear directions (Azevedo et al., 2026; Brockhoff & Zitzler, 2006). Because NBI constructs the CHIM from the individual objective optima and defines its search directions from this geometry (Das & Dennis, 1998), such dependencies may lead to rank deficiency or poor conditioning in the matrix representing the CHIM geometry. From a numerical linear algebra perspective, near-linear dependence is characterized by small singular values and increased sensitivity of the associated subspace and directional calculations (Golub & Van Loan, 2013).

Therefore, simply selecting subsets containing no more than $n_x + 1$ objectives resolves dimensional admissibility but does not guarantee that the retained objectives provide sufficient independent geometric information for a meaningful NBI search. Extending NBI to high-dimensional multiobjective problems, particularly when the number of objectives exceeds the dimensional capacity of the direct formulation, therefore requires addressing both dimensional admissibility and geometric informativeness.

Many-objective evolutionary algorithms approach this problem from a different direction. As the number of objectives increases, however, Pareto-based selection becomes progressively less discriminative because a growing proportion of the population may become mutually nondominated, a phenomenon associated with dominance resistance and reduced selection pressure (Purshouse & Fleming, 2007). NSGA-III uses reference points to maintain diversity among nondominated solutions (Deb & Jain, 2014), whereas MOEA/D decomposes a multiobjective problem into

coordinated scalar subproblems (Q. Zhang & Li, 2007). More recent methods adapt dominance relations, reference vectors, and decomposition structures to improve search behavior as objective dimensionality increases (Bao et al., 2023; W. Zhang et al., 2023). These developments demonstrate the flexibility of evolutionary optimization in high-dimensional objective spaces. At the same time, convergence assessment, diversity preservation, reference-set design, and computational effort become increasingly challenging as the number of objectives grows. An NBI-based alternative therefore remains valuable when the explicit geometric relationship between individual optima and generated compromises is desirable.

Another strategy is to reduce the objective system before optimization. Principal component analysis constructs lower-dimensional representations through orthogonal linear combinations of the original variables that successively maximize explained variance (Jolliffe & Cadima, 2016), whereas factor-based approaches similarly seek lower-dimensional latent structures. In engineering applications, these techniques have been combined with NBI to restore compatibility with problems in which the original number of responses exceeds the dimensional capacity of the direct formulation (Belinato et al., 2019; Naves et al., 2017; Pereira & Paiva, 2025; Streitenberger et al., 2022). The Varimax-Rotated Factor Normal Boundary Intersection (VRF-NBI), for example, combines factor analysis, Varimax rotation, and NBI to optimize systems with numerous correlated responses (Pereira et al., 2025). This strategy can reduce redundancy and produce a compact representation of the optimization problem. However, the Pareto search is performed in a transformed latent space. Although the original objectives can subsequently be recovered, they no longer constitute the search dimensions that directly generate the compromise solutions.

Objective reduction provides another way of controlling objective-space dimensionality. Such approaches identify objectives considered redundant or nonessential and remove them from the optimization problem, thereby simplifying the subsequent search (Sinha et al., 2013). This strategy differs fundamentally from the mechanism considered here: an objective excluded from a CNBI subset is omitted only temporarily during candidate generation and remains part of the complete problem during final evaluation.

These observations expose a broader methodological gap. To the authors' knowledge, existing NBI formulations have not combined temporary objective-subset search, geometry-based subset screening, and final evaluation in the complete original objective

space. The research question addressed here is therefore: how can NBI explore structurally dependent high-dimensional multiobjective problems while preserving the original objective meanings and avoiding dimensionally or geometrically unsuitable subproblems?

This study addresses this question by proposing Combinatorial Normal Boundary Intersection (CNBI). Rather than replacing the original objectives with latent variables, CNBI decomposes the m -objective problem into temporary subsets of cardinality k , satisfying $2 \leq k \leq n_x + 1$, so that the corresponding NBI subproblems remain dimensionally admissible. Each subset is used only as a candidate-generation problem; it does not replace the original optimization problem.

Dimensional admissibility alone, however, does not ensure that a subset provides a useful NBI geometry. CNBI therefore incorporates a spectral screening procedure based on the singular-value structure of the CHIM associated with each candidate subset. The procedure evaluates whether the corresponding anchors contain sufficient independent geometric structure and excludes combinations characterized by excessive redundancy or degeneracy before optimization. The screening limits are derived from information contained in the global payoff geometry and from parallel analysis (Horn, 1965). When the objective functions originate from fitted statistical models, model uncertainty can additionally be incorporated into this calibration. In the RSM application considered in this study, parallel analysis is calibrated using response-surface prediction uncertainty. This uncertainty-aware implementation constitutes an application-specific component of the screening procedure rather than a restriction of the CNBI decomposition principle itself.

NBI is then solved for each retained subset. The resulting decision vectors are reevaluated using all m original objective functions, merged, deduplicated, and filtered by global Pareto dominance. Lower-dimensional objective subsets are therefore used exclusively during candidate generation. Final acceptance, Pareto evaluation, and interpretation remain in the complete original objective space, so that no objective is permanently discarded, aggregated, or replaced by a latent variable.

The methodological contribution is threefold. First, CNBI introduces a combinatorial decomposition mechanism that extends normal-boundary candidate generation to problems for which direct NBI is dimensionally unsuitable. Second, it combines dimensional admissibility with spectral screening to distinguish subsets that can merely be solved from those that contain sufficient geometric information to justify an NBI

search. Third, it separates the space used for candidate generation from the space used for final evaluation, preserving all original objectives while exploiting lower-dimensional searches during optimization.

The evaluation was designed to distinguish controlled evidence, practical evidence, and stress testing. A synthetic experiment comprises 27 structural scenarios and 10 replications, varying decision-space dimension, dimensional excess, and dependence among objectives. CNBI is compared with VRF-NBI, NSGA-III, and MOEA/D. Complete fronts constitute the primary comparison, while fixed-cardinality analyses from 2 to 20 solutions evaluate the sensitivity of the conclusions to the number of returned alternatives. A nine-scenario ablation study compares spectral screening with the exhaustive solution of all dimensionally admissible subsets. Practical evidence is provided by an eight-response RSM application to hardened AISI H13 steel turning with PCBN 7025 inserts. Finally, MaF8, MaF9, and MaF13 benchmark suite (Cheng et al., 2017) are used as complementary CNBI stress tests outside the response-surface setting, providing evidence of its behavior and limitations beyond DOE/RSM-based models.

The remainder of this paper is organized as follows. Section 2 presents the theoretical basis of CNBI and its spectral screening procedure. Section 3 describes the synthetic experimental design, comparison methods, performance indicators, manufacturing application, stress tests, and statistical analyses. Section 4 reports the results, Section 5 discusses their methodological and practical implications, Section 6 addresses the limitations of the proposed framework, and Section 7 presents the conclusions.

2. Combinatorial normal boundary intersection

This section defines the mathematical formulation and computational sequence of combinatorial normal boundary intersection (CNBI). The central distinction is between the objective space used to generate candidates and the objective space used to accept them. Dimensionally admissible subsets support local NBI searches, whereas every resulting decision is evaluated and filtered using the complete set of original objectives. Thus, the decomposition changes the search mechanism without redefining the decision problem.

2.1 Problem formulation and normalized payoff geometry

Consider a continuous multiobjective minimization problem with n_x decision variables and m objective functions. The feasible region Ω includes all variable bounds and problem constraints. Objectives originally defined for maximization are multiplied by -1 before the optimization, so that a common minimization convention is used throughout the method.

$$\min_{(x \in \Omega)} F(x) = [f_1(x), \dots, f_m(x)]^T, x \in R(n_x) \quad (1)$$

For each objective f_j , an anchor decision x_j^* is obtained by individual optimization. Evaluating all objectives at every anchor produces the payoff matrix Φ . Its j th column is the complete objective vector generated by the individual optimum of f_j . Off-diagonal entries are retained because they describe the deterioration of the other objectives when f_j is optimized alone (Das & Dennis, 1998).

$$x_j^* \in \operatorname{argmin}_{(x \in \Omega)} f_j(x), \Phi_{(ij)} = f_i(x_j^*), i, j = 1, \dots, m \quad (2)$$

Objective scales can differ by several orders of magnitude. CNBI therefore uses a common affine transformation based on the ideal value f_i^U and the amplitude a_i represented in the payoff geometry. For ordinary RSM problems, f_i^U is the row minimum of Φ and a_i is the difference between its maximum and minimum. Benchmark functions with known theoretical scales use the declared ideal and amplitude. The same transformation is applied to the payoff matrix and to every objective vector used inside an NBI subproblem.

$$f_i(x) = (f_i(x) - f_i^U)/a_i, a_i = f_i^N - f_i^U > 0 \quad (3)$$

Let Φ contain the normalized payoff columns. Their convex hull of individual minima (CHIM) is parameterized by a vector β in the unit simplex. The weights select geometric points on the CHIM; they are not preference coefficients used to aggregate the objective functions.

$$CHIM = \Phi\beta, \beta \in \Delta_{(m-1)}; \Delta_{(m-1)}: \beta \in R^m, \beta_i \geq 0, \sum_{(i=1)}^m \beta_i = 1 \quad (4)$$

2.2 Direct NBI and dimensional compatibility

For a fixed β , direct NBI searches from the corresponding CHIM point in an outward normal direction n . The displacement t is maximized until the normalized attainable objective boundary is reached. The equality form used in this study is given in Eq. (5), with feasibility enforced through $x \in \Omega$.

$$\max_{(x,t)} t: F(x) - \Phi\beta - tn = 0, x \in \Omega \quad (5)$$

Equation (5) contains m scalar equalities. Once β is fixed, its unknowns are the n_x decision variables and the scalar t , totaling $n_x + 1$ unknowns. Ignoring special dependencies among equations, a necessary counting condition for the system not to be generically overdetermined is therefore given by Eq. (6).

$$m \leq n_x + 1 \quad (6)$$

When $m > n_x + 1$, a direct solution can occur only under particular dependencies in the objective mapping. This dimensional restriction is separate from statistical correlation. Even when Eq. (6) is satisfied, anchor points can be linearly or nearly linearly dependent, causing the CHIM to collapse or become poorly conditioned. CNBI addresses the equation-count restriction through temporary decomposition and addresses weak geometry through spectral screening.

2.3 Temporary combinatorial decomposition

Let $I=\{1,\dots,m\}$ denote the complete objective index set. CNBI enumerates all unordered subsets $S\subseteq I$ whose cardinality k is at least two and does not exceed either m or the direct NBI capacity $n_x + 1$. The candidate family is defined by Eq. (7).

$$\mathcal{C} = \bigcup_{(k=2)}^{(\min(m,n_x+1))} \{S \subseteq I: |S| = k\} \quad (7)$$

The number of candidate subsets follows directly from the binomial coefficients. This quantity can grow rapidly with m , which motivates screening before local optimization.

$$|\mathcal{C}| = \sum_{(k=2)}^{(\min(m,n_x+1))} C(m, k) \quad (8)$$

For each S , CNBI extracts from the global normalized payoff matrix the rows and anchor columns associated with the same original objectives. The resulting $k \times k$ payoff submatrix defines a local CHIM. The subset is temporary: it specifies which equalities are active in a local NBI search but does not remove the remaining objectives from the final problem. Since an objective can participate in several subsets, the union of local searches explores different projections of the original trade-off structure. This differs from objective-reduction methods that permanently omit responses or replace them with latent variables (Brockhoff & Zitzler, 2006).

2.4 Spectral representation of payoff geometry

Dimensional admissibility does not guarantee that a local CHIM has the expected geometric dimension. Let $A_s=[a_1^T, \dots, a_k^T]^T$ be the normalized anchor matrix for subset S , with one anchor in each row. Taking the first anchor as reference produces the edge matrix E_s .

$$E_s = [(a_2 - a_1)^T; \dots; (a_k - a_1)^T] \in R^{(k-1) \times k} \quad (9)$$

The singular value decomposition separates the orthogonal directions of the local geometry and their magnitudes (Golub & Van Loan, 2013). A full-dimensional local simplex requires $\text{rank}(E_s)=k-1$. An exactly deficient matrix is rejected because its normal direction is not unique in the form required by the local NBI construction.

$$E_s = U_s \Sigma_s V_s^T, \sigma_1(S) \geq \dots \geq \sigma_{(k-1)}(S) \geq 0 \quad (10)$$

The smallest singular value measures the weakest direction of the local CHIM. A small value indicates that the anchors approach a lower-dimensional configuration. Conditioning measures the relative imbalance between the strongest and weakest directions. A large condition number characterizes an elongated geometry in which uniform changes in simplex weights can correspond to very different displacements in objective space.

$$q_s = \sigma_1(S)/\sigma_{(k-1)}(S) \quad (11)$$

Neither the smallest singular value nor q_s has a universal acceptable threshold. CNBI derives both limits from the normalized global payoff geometry, allowing the criterion to

adapt to the structural scale of each problem rather than relying on an arbitrary fixed tolerance. The contrast is illustrated in Fig. 1: an additive shift produces no new trade-off direction, whereas genuinely conflicting objectives define a nondegenerate local geometry.

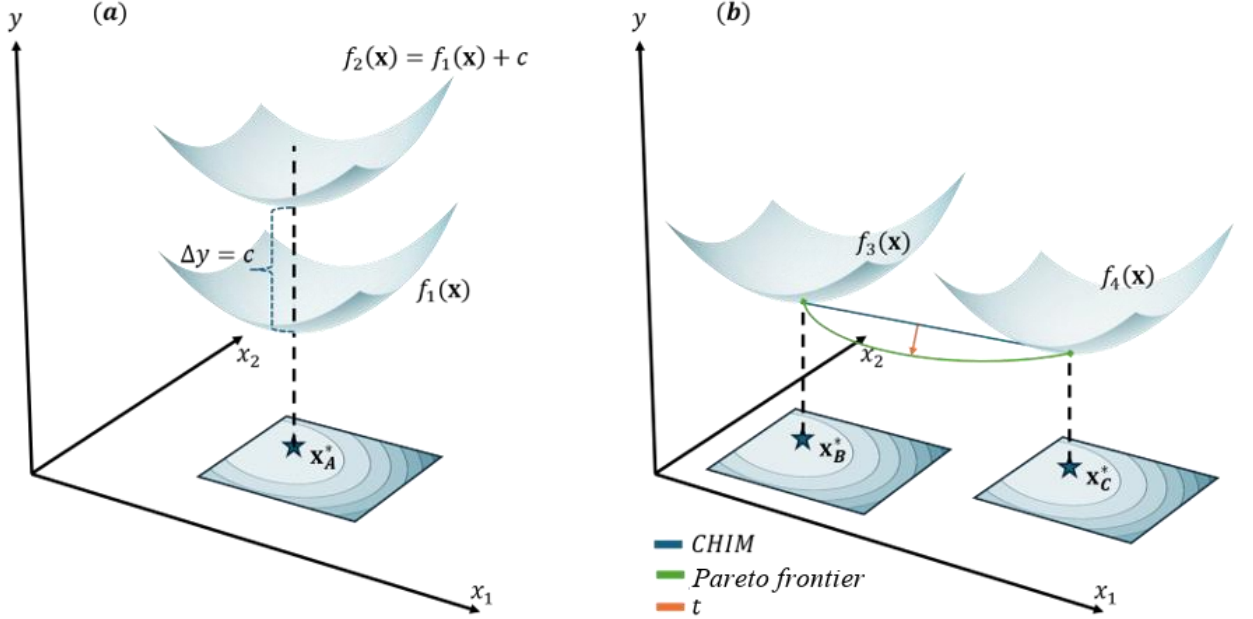


Fig. 1. Geometric motivation for the spectral screening used by CNBI. (a) Objectives related by an additive constant generate redundant payoff directions. (b) Conflicting objectives define a nondegenerate CHIM direction (blue), a Pareto-front segment (green), and the NBI displacement t (orange).

2.5 Effective dimension and noise-aware calibration

The global edge matrix E_G is constructed from all normalized payoff anchors and decomposed by SVD. In RSM applications, the anchors are themselves obtained from estimated models. Small observed directions can therefore reflect either real trade-off structure or prediction uncertainty. CNBI adapts Horn's parallel-analysis principle to distinguish these possibilities (Horn, 1965).

For objective j at anchor decision $\mathbf{x}_i^*$, the standard error of the estimated mean response is calculated from the residual mean square MSE_j , the model vector $\mathbf{z}(\mathbf{x}_i^*)$, and the inverse information matrix. Division by a_j places this uncertainty on the same normalized scale as the payoff geometry (Myers et al., 2016).

$$s_{(ij)} = \sqrt{(MSE_j \mathbf{z}(\mathbf{x}_i^*)^T (X^T X)^{(-1)} \mathbf{z}(\mathbf{x}_i^*))} / a_j. \quad (12)$$

CNBI generates $B=2,000$ independent Gaussian perturbation matrices with element-specific standard deviations s_{ij} . Each realization is converted to an edge matrix and decomposed by SVD. For spectral position r , τ_r is the 95th percentile of the corresponding simulated singular values.

$$\varepsilon_{(ij)}^{(b)} \sim N(0, s_{(ij)}^2), \tau_r = Q_{(0.95)}\{\sigma_r(E_\varepsilon^{(b)}) : b = 1, \dots, 2000\} \quad (13)$$

The effective dimension d is the number of observed global directions that exceed their position-specific noise thresholds. The implementation retains a minimum operational dimension of one when no observed direction exceeds its threshold. This lower bound prevents a zero-dimensional diagnostic; it does not restrict CNBI to subsets with only d objectives.

$$d = \max(1, \sum_r 1[\sigma_r(E_G) > \tau_r]) \quad (14)$$

The benchmark stress tests do not supply fitted-model prediction variances. For MaF8, MaF9, and MaF13, CNBI therefore uses a separate nonparametric calibration: $N=500$ reference-front objective vectors are standardized, each objective column is independently permuted $B=1,000$ times, and the observed eigenvalue spectrum is compared sequentially with the 95th-percentile null spectrum. This permutation procedure estimates d for the benchmark diagnostic only; it does not alter the uncertainty-based RSM procedure used in the synthetic DOE and turning application.

The effective dimension is converted into a global spectral window. The first discarded global direction supplies the magnitude floor, while the condition number of the retained global subspace supplies the maximum admissible anisotropy. When every available direction is retained, the floor is zero.

$$\tau_\sigma = \sigma_{(d+1)}(E_G), d < \text{rank}(E_G); \tau_\sigma = 0, d = \text{rank}(E_G); \tau_q = \sigma_1(E_G)/\sigma_d(E_G) \quad (15)$$

2.6 Spectral selection and local NBI solution

A dimensionally admissible subset proceeds to optimization only if its weakest direction exceeds the global residual floor and its condition number does not exceed that of the retained global subspace. The joint rule in Eq. (16) rejects geometries that are too thin in absolute terms, too anisotropic in relative terms, or exactly rank deficient.

$$\text{retain } S \Leftrightarrow \text{rank}(E_S) = k - 1, \sigma_{(k-1)}(S) > \tau_\sigma, q_S \leq \tau_q. \quad (16)$$

This window is a heuristic screening rule derived from the problem's own geometry. It is not claimed as a necessary or sufficient condition for NBI convergence. Its purpose is to avoid spending optimization effort on sub-CHIMs whose normal directions are poorly supported by the global structure or indistinguishable from the uncertainty scale.

For every retained subset, weight vectors are generated by a Simplex-Lattice design. Let δ_k denote the lattice resolution and $H_k=1/\delta_k$ the number of divisions. The final campaign uses $\delta_k=0.20$ for $k \leq 4$ and $\delta_k=0.50$ for $k > 4$. Thus, smaller subsets receive a finer geometric search, whereas higher-dimensional subsets use a coarser lattice to control combinatorial growth.

$$B_k: \beta \in \Delta_{(k-1)}, H_k \beta_i \in N_0, H_k = 1/\delta_k. \quad (17)$$

The number of local NBI subproblems contributed by one retained subset is given by Eq. (18). Consequently, the spectral filter affects cost twice: it removes entire subsets and all lattice points associated with them.

$$|B_k| = C(H_k + k - 1, k - 1) \quad (18)$$

For a retained subset S and weight $\beta \in B_k$, the normalized local NBI problem uses the local anchor matrix A_s and its outward normal n_s . The implementation minimizes $-t$, which is equivalent to maximizing t in Eq. (19).

$$\max_{(x,t)} t: F_S(x) - \beta^T A_S - t n_s = 0, x \in \Omega \quad (19)$$

The local problems are solved by sequential least-squares programming. For quadratic RSM objectives, analytical objective gradients and equality Jacobians are used. For general benchmark functions, finite-difference Jacobians are fully counted as objective-vector evaluations. Exact anchor weights reuse the corresponding individual-optimum decisions. Interior weights use the preceding accepted solution, the convex combination of anchor decisions, anchor decisions ordered by weight, the domain center, and reproducibly generated rescue points as candidate starts. A subproblem is accepted only when the decision is feasible and the infinity norm of the NBI equality residual does not exceed 10^{-5} . The final solver configuration allows 100 iterations per start and uses an internal optimization tolerance of 10^{-10} .

2.7 Recomposition in the complete objective space

All accepted local decisions are pooled after the subset searches. Each decision is then reevaluated using all m original objective functions supplied to the algorithm, regardless of the subset that generated it. This step restores the complete decision problem before any final quality judgment is made.

Post-processing first removes infeasible decisions. Decisions whose coordinates coincide after quantization at 10^{-5} are treated as duplicates; within each group, the candidate with the smallest sum of normalized minimization objectives is retained. Global Pareto filtering is then applied in the complete m -dimensional objective space with a numerical dominance tolerance of 10^{-10} . A point generated by a locally efficient subset can therefore be removed when another candidate dominates it after all objectives are considered.

$$P_C NBI = ND_m(F(x): x \in \bigcup_{(S \in C_{retain})} X_S) \quad (20)$$

CNBI returns the complete nondominated set as its primary output. It also records the generating subset, lattice weight, convergence status, equality residual, feasibility result, and number of attempts for each candidate. This audit trail makes it possible to distinguish screening decisions, numerical failures, duplicate generation, and removal by global dominance. Algorithm 1 summarizes the complete procedure.

Algorithm 1. Combinatorial normal boundary intersection

Step	Operation
1	Orient all objectives to minimization and determine each individual optimum.
2	Construct and normalize the global payoff matrix and its anchor decisions.

3	Estimate the effective geometric dimension and the global spectral window.
4	Enumerate every objective subset with $2 \leq k \leq \min(m, n_x + 1)$.
5	Build the local edge matrix and reject exactly degenerate subsets.
6	Retain a subset only when its weakest direction and conditioning satisfy the spectral window.
7	Generate Simplex-Lattice weights using the resolution assigned to k .
8	Solve the local NBI subproblems and retain feasible converged decisions.
9	Reevaluate every retained decision in all m original objectives.
10	Remove duplicate decisions and apply global nondominance filtering.
11	Return the final frontier and an audit ledger linking solutions to subsets and weights.

The worst-case number of local subproblems before screening is the sum, over all admissible cardinalities, of the number of objective subsets multiplied by the corresponding lattice size. The realized cost is lower when the spectral window rejects combinations. Because each nonlinear subproblem can require a different number of objective and gradient evaluations, the experimental comparison reports measured objective-vector evaluations rather than relying only on the combinatorial upper bound.

Funding

The authors declare that they have no relevant financial or non-financial support to disclose.

Credit authorship contribution statement

G. V. L.: Methodology, Software, Visualization, Formal analysis, Writing – review & editing. **M. C. C.:** Writing – review & editing. **A. P. P.:** Supervision, Resources, Writing - Review & Editing.

Data availability

Declaration of Competing Interests

The authors declare no competing interests.

Consent to Participate and Publication

All authors provided informed consent to participate in this study. All authors gave their consent for the publication of this manuscript.

Acknowledgements

The authors would like to thank Federal University of Itajubá (UNIFEI) and Nucleus of Manufacturing Optimization and Innovation Technology (NOMATI) for providing access to laboratories, materials, and technical expertise.

References

- Allmendinger, R., Jaszkievicz, A., Liefoghe, A., & Tammer, C. (2022). What if we increase the number of objectives? Theoretical and empirical implications for many-objective combinatorial optimization. *Computers & Operations Research*, 145, 105857. <https://doi.org/10.1016/j.cor.2022.105857>
- Azevedo, T. M. D., Pereira, M. C., Cesário, M. D. C., & Paiva, A. P. D. (2026). Multiobjective and multivariate rationalization of CFD simulations for hydrodynamic systems using the NBI-VRF method. *Thermal Science and Engineering Progress*, 74, 104722. <https://doi.org/10.1016/j.tsep.2026.104722>
- Bao, C., Gao, D., Gu, W., Xu, L., & D. Goodman, E. (2023). A new adaptive decomposition-based evolutionary algorithm for multi- and many-objective optimization. *Expert Systems with Applications*, 213, 119080. <https://doi.org/10.1016/j.eswa.2022.119080>
- Belinato, G., De Almeida, F. A., De Paiva, A. P., De Freitas Gomes, J. H., Balestrassi, P. P., & Rosa, P. A. R. C. (2019). A multivariate normal boundary intersection PCA-based approach to reduce dimensionality in optimization problems for LBM process. *Engineering with Computers*, 35(4), 1533–1544. <https://doi.org/10.1007/s00366-018-0678-3>
- Brockhoff, D., & Zitzler, E. (2006). Are All Objectives Necessary? On Dimensionality Reduction in Evolutionary Multiobjective Optimization. In T. P. Runarsson, H.-G. Beyer, E. Burke, J. J. Merelo-Guervós, L. D. Whitley, & X. Yao (Eds.), *Parallel Problem Solving from Nature—PPSN IX* (Vol. 4193, pp. 533–542). Springer Berlin Heidelberg https://doi.org/10.1007/11844297_54

- Cheng, R., Li, M., Tian, Y., Zhang, X., Yang, S., Jin, Y., & Yao, X. (2017). A benchmark test suite for evolutionary many-objective optimization. *Complex & Intelligent Systems*, 3(1), 67–81. <https://doi.org/10.1007/s40747-017-0039-7>
- Chiao, C.-H., & Hamada, M. (2001). Analyzing Experiments with Correlated Multiple Responses. *Journal of Quality Technology*, 33(4), 451–465. <https://doi.org/10.1080/00224065.2001.11980104>
- Das, I., & Dennis, J. E. (1997). A closer look at drawbacks of minimizing weighted sums of objectives for Pareto set generation in multicriteria optimization problems. *Structural Optimization*, 14(1), 63–69. <https://doi.org/10.1007/BF01197559>
- Das, I., & Dennis, J. E. (1998). Normal-Boundary Intersection: A New Method for Generating the Pareto Surface in Nonlinear Multicriteria Optimization Problems. *SIAM Journal on Optimization*, 8(3), 631–657. <https://doi.org/10.1137/S1052623496307510>
- Deb, K., & Jain, H. (2014). An Evolutionary Many-Objective Optimization Algorithm Using Reference-Point-Based Nondominated Sorting Approach, Part I: Solving Problems With Box Constraints. *IEEE Transactions on Evolutionary Computation*, 18(4), 577–601. <https://doi.org/10.1109/TEVC.2013.2281535>
- Golub, G. H., & Van Loan, C. F. (2013). *Matrix computations* (4. ed). Johns Hopkins Univ. Press.
- Horn, J. L. (1965). A Rationale and Test for the Number of Factors in Factor Analysis. *Psychometrika*, 30(2), 179–185. <https://doi.org/10.1007/BF02289447>
- Jeong, I.-J., Lee, D.-H., & Son, Y.-J. (2024). A process capability function approach to multiple response surface optimization based on a posterior procedure. *Quality Engineering*, 36(4), 831–844. <https://doi.org/10.1080/08982112.2024.2310264>
- Jolliffe, I. T., & Cadima, J. (2016). Principal component analysis: A review and recent developments. *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences*, 374(2065), 20150202. <https://doi.org/10.1098/rsta.2015.0202>

- Kumar, R., & Reji, M. (2023). Response surface methodology (RSM): An overview to analyze multivariate data. *Indian Journal of Microbiology Research*, 9(4), 241–248. <https://doi.org/10.18231/j.ijmr.2022.042>
- Mechaacha, A., Belkaid, F., & Brahimi, N. (2026). Multi-objective multi-product process planning with reconfigurable machines: Exact and metaheuristic approaches. *Expert Systems with Applications*, 298, 129591. <https://doi.org/10.1016/j.eswa.2025.129591>
- Montgomery, D. C. (2013). *Design and analysis of experiments* (Eighth edition). John Wiley & Sons, Inc.
- Moura, D., Barcelos, V., Samanamud, G. R. L., França, A. B., Lofrano, R., Loures, C. C. A., Naves, L. L. R., Amaral, M. S., & Naves, F. L. (2018). Normal boundary intersection applied as multivariate and multiobjective optimization in the treatment of amoxicillin synthetic solution. *Environmental Monitoring and Assessment*, 190(3), 140. <https://doi.org/10.1007/s10661-018-6523-8>
- Myers, R. H., Montgomery, D. C., & Anderson-Cook, C. M. (2016). *Response surface methodology: Process and product optimization using designed experiments* (4th ed.). John Wiley & Sons.
- Naves, F. L., De Paula, T. I., Balestrassi, P. P., Moreira Braga, W. L., Sawhney, R. S., & De Paiva, A. P. (2017). Multivariate Normal Boundary Intersection based on rotated factor scores: A multiobjective optimization method for methyl orange treatment. *Journal of Cleaner Production*, 143, 413–439. <https://doi.org/10.1016/j.jclepro.2016.12.092>
- Pereira, M. C., & Paiva, A. P. de. (2025). Nonlinear multiobjective optimization of efficiency conditions using a CFD-DOE hybrid approach: A practical application in centrifugal fans for industrial ovens. *Thermal Science and Engineering Progress*, 103899. <https://doi.org/10.1016/j.tsep.2025.103899>
- Pereira, M. C., Ribeiro, C. T., & Mendes, R. R. (2025). A hybrid multivariate normal boundary intersection approach with post-optimization assisted by mixture design of experiments. *Engineering Applications of Artificial Intelligence*, 112510. <https://doi.org/10.1016/j.engappai.2025.112510>

- Purshouse, R. C., & Fleming, P. J. (2007). On the Evolutionary Optimization of Many Conflicting Objectives. *IEEE Transactions on Evolutionary Computation*, 11(6), 770–784.
<https://doi.org/10.1109/TEVC.2007.910138>
- Sinha, A., Saxena, D. K., Deb, K., & Tiwari, A. (2013). Using objective reduction and interactive procedure to handle many-objective optimization problems. *Applied Soft Computing*, 13(1), 415–427. <https://doi.org/10.1016/j.asoc.2012.08.030>
- Streitenberger, S. C., Romão, E. L., Paiva, A. P. de, Balestrassi, P. P., Freitas, J. H. G., & Paes, V. C. (2022). Normal boundary intersection with factor analysis approach for multiobjective stochastic optimization of a cladding process focusing on reduction of energy consumption and rework. *Journal of Cleaner Production*, 333, 130269.
<https://doi.org/10.1016/j.jclepro.2021.129915>
- Tong, L.-I., Wang, C.-H., & Chen, H.-C. (2005). Optimization of multiple responses using principal component analysis and technique for order preference by similarity to ideal solution. *The International Journal of Advanced Manufacturing Technology*, 27(3–4), 407–414. <https://doi.org/10.1007/s00170-004-2157-9>
- Zhang, Q., & Li, H. (2007). MOEA/D: A Multiobjective Evolutionary Algorithm Based on Decomposition. *IEEE Transactions on Evolutionary Computation*, 11(6), 712–731.
<https://doi.org/10.1109/TEVC.2007.892759>
- Zhang, W., Liu, J., Tan, S., & Wang, H. (2023). A decomposition-rotation dominance based evolutionary algorithm with reference point adaption for many-objective optimization. *Expert Systems with Applications*, 215, 119424.
<https://doi.org/10.1016/j.eswa.2022.119424>